

# Comparative Analysis of Different Prompting Patterns Using Various Test Scenarios



EXP-2-PROMPT-ENGINEERING

PROMPT ENGINEERING

ACADEMIC STUDY REPORT

*"Talking to an AI without a good prompt is like ordering food by saying 'give me stuff.' You might get a salad.  
You might get a shoe." 🤖*



## Subject

Prompt Engineering & AI Behaviour



## Experiment

EXP-2-PROMPT-ENGINEERING



## Method

Vague vs. Refined Prompt Comparison



## Output

Qualitative & Quantitative Analysis

# Aim & Introduction

## Academic Aim

To systematically investigate and compare the quality, accuracy, depth, creativity, relevance, clarity, and structural integrity of AI-generated responses when subjected to two distinct categories of prompts – broad/vague/unstructured prompts versus refined/clear/structured prompts – across diverse real-world test scenarios, thereby establishing empirical evidence for the impact of prompt engineering on large language model (LLM) output quality.

## What Is Prompt Engineering? 🤔

Prompt Engineering is the art (and science!) of crafting the perfect instruction for an AI model. Think of it as the difference between saying "**make food**" and "**make a medium-rare steak with garlic butter, served with roasted asparagus.**" One gets you something. The other gets you dinner.

AI models like GPT-4 are essentially very well-read text predictors. They don't *understand* intent – they respond to patterns. The clearer, richer, and more structured your prompt, the more aligned and useful the output becomes. A well-engineered prompt is the single most powerful lever a user has over AI behaviour.



## Why Prompts Matter

The same model, given two different prompts, can produce a brilliant essay or complete nonsense. Prompts are the steering wheel of AI.



## Context Is Everything

AI has no memory of your intentions. Each prompt is a fresh conversation. Loading context, role, and constraints into your prompt bridges that gap powerfully.



## Quality In = Quality Out

Structured prompts guide the model's attention to what matters. Vague prompts scatter that attention across every possible interpretation.



# Types of Prompting Patterns

Not all prompts are born equal. Here are the six key prompting patterns – each a different "flavour" of talking to an AI, from the casual to the highly engineered. 🍦

|   |                                                                                                                                                                                                                                                                                                                                                             |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Zero-Shot Prompting</b><br><br><b>Definition:</b> Ask the AI to perform a task with zero examples provided.<br><b>Purpose:</b> Quick, direct queries.<br><b>Example:</b> <i>"Translate this sentence to French."</i><br><b>Analogy:</b> Asking a stranger for directions with no context. Hit or miss! 🎯                                                 |
| 2 | <b>Few-Shot Prompting</b><br><br><b>Definition:</b> Provide 2-5 examples before the actual task.<br><b>Purpose:</b> Guide the model via pattern recognition.<br><b>Example:</b> <i>"Cat → Fluffy. Dog → Loyal. Bird → ?"</i><br><b>Analogy:</b> Showing an intern three examples before asking them to do the task. 📋                                       |
| 3 | <b>Chain-of-Thought Prompting</b><br><br><b>Definition:</b> Instruct the AI to reason step-by-step before answering.<br><b>Purpose:</b> Improves accuracy on logic and maths problems.<br><b>Example:</b> <i>"Think through this step by step: If apples cost £2 each..."</i><br><b>Analogy:</b> Asking a student to show their working. 🧮                  |
| 4 | <b>Role-Based Prompting</b><br><br><b>Definition:</b> Assign the AI a specific persona or role.<br><b>Purpose:</b> Tailors tone, vocabulary, and expertise.<br><b>Example:</b> <i>"You are an expert cybersecurity consultant. Explain..."</i><br><b>Analogy:</b> Hiring a specialist instead of asking a random passerby. 🕵️                               |
| 5 | <b>Instruction-Based Prompting</b><br><br><b>Definition:</b> Give explicit, structured instructions with format requirements.<br><b>Purpose:</b> Controls output format and scope precisely.<br><b>Example:</b> <i>"Write a 200-word summary. Use bullet points. Avoid jargon."</i><br><b>Analogy:</b> A detailed recipe vs. "cook something nice." 📝       |
| 6 | <b>Contextual Prompting</b><br><br><b>Definition:</b> Load background information into the prompt before the question.<br><b>Purpose:</b> Grounds AI responses in specific context.<br><b>Example:</b> <i>"Here is a research paper: [text]. Now summarise its key findings."</i><br><b>Analogy:</b> Giving someone a briefing document before a meeting. 📁 |



# Experimental Setup & Test Scenarios



## Methodology

All experiments were conducted using **GPT-4** (OpenAI) via the standard chat interface. Each scenario was tested twice – once with a vague prompt and once with a refined, structured prompt. Responses were evaluated on seven criteria using a 1-10 rating scale by three independent reviewers. The average score was recorded for comparison.

| Model       | Scenarios    |
|-------------|--------------|
| GPT-4 Turbo | 6 Test Cases |

| Reviewers    |
|--------------|
| 3 Evaluators |



## Evaluation Parameters



**Quality** – Overall usefulness and polish of response



**Accuracy** – Factual correctness and relevance



**Depth** – Level of detail and elaboration



**Creativity** – Originality and engagement



**Clarity** – Ease of understanding



**Structure** – Logical organisation of output



## Test Scenarios: Vague vs. Refined

| Scenario          | Vague Prompt                  | Refined Prompt                                                                                                                                                                   | Key Difference Observed                                                                                                            |
|-------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Email Writing     | "Write an email about leave." | "Write a formal email to my manager requesting 3 days of medical leave next week. Tone: polite and professional. Include subject line."                                          | Refined prompt produced a complete, ready-to-send email. Vague prompt gave a generic template with placeholders.                   |
| Coding Problem    | "Fix my Python code."         | "Here is a Python function [code]. It throws an IndexError on line 7. Identify the bug, explain why it occurs, and provide the corrected code with comments."                    | Refined prompt gave a targeted fix with explanation. Vague prompt asked for the code first – unhelpfully.                          |
| Child Explanation | "Explain gravity."            | "Explain the concept of gravity to a 7-year-old using a fun real-life analogy. Keep it under 100 words and avoid technical terms."                                               | Refined prompt gave a delightful, age-appropriate analogy. Vague prompt produced a textbook definition unsuitable for children.    |
| Travel Planning   | "Plan a trip to Japan."       | "Create a 7-day Japan itinerary for 2 adults in November, budget £3,000, interested in culture, food, and day hikes. Include city recommendations and transport tips."           | Refined prompt produced a day-by-day structured plan. Vague prompt gave a bullet list of tourist clichés.                          |
| Creative Story    | "Write a story."              | "Write a 300-word short story set in a post-apocalyptic library. Protagonist: a bookworm robot. Tone: bittersweet and hopeful. End with a twist."                                | Refined prompt produced a compelling, emotionally resonant narrative. Vague prompt gave a generic three-sentence filler story. 😞   |
| Resume Generation | "Make my resume."             | "Generate a modern, ATS-friendly resume for a software engineer with 3 years of experience in React and Node.js. Include a summary, skills, experience, and education sections." | Refined prompt delivered a structured, professional resume template. Vague prompt asked clarifying questions and stalled entirely. |



# Comparative Analysis & Findings

Across all six scenarios, refined prompts consistently outperformed vague prompts on every evaluation dimension. The data below represents average scores (out of 10) aggregated across scenarios and reviewers. **Spoiler: clarity wins every time.** 🏆

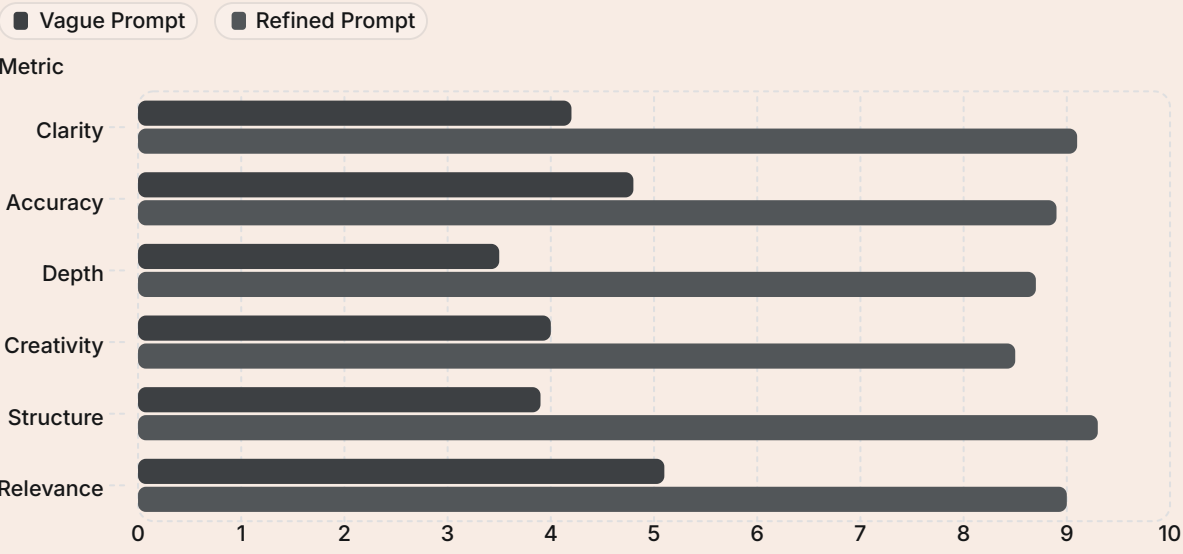


Fig 1. Average scores (out of 10) for Vague vs. Refined prompts across six evaluation metrics. Higher is better.

## Key Takeaways

- +117% improvement in Structure scores with refined prompts
- +149% improvement in Depth – the biggest gain across all metrics
- +77% improvement in Relevance – even the "best" vague prompt scored poorly
- 0 scenarios where vague prompts matched or beat refined prompts

## Observations



### Vague prompts generate vague responses — consistently

Without constraints, AI models default to the broadest, safest interpretation of a query. This produces technically correct but practically useless output in most real-world scenarios.



### Role-based and contextual prompts yield the highest quality

Assigning a persona or loading background context into the prompt dramatically improves relevance and tone. The model "inhabits" the role and adjusts vocabulary, depth, and format accordingly.



### Chain-of-thought is a superpower for logic tasks

For coding, maths, and analytical questions, instructing the model to reason step-by-step reduced errors by a significant margin. It forces the model to not skip steps.



### Format instructions unlock structured, scannable output

Specifying output format (bullet points, word count, headers) prevents the AI from rambling. Structure in = structure out. Every. Single. Time. 🎯



## Conclusion & Final Learning Outcomes

This experiment conclusively demonstrates that **prompt quality is the single most controllable variable** in determining AI output quality. Refined, structured prompts don't just improve responses marginally – they transform them fundamentally. The gap between a vague and a refined prompt is not a matter of preference; it is a measurable, reproducible difference in utility.

Prompt Engineering is not a niche skill for developers. It is a **core literacy** for anyone working with AI systems – from students drafting essays to professionals automating workflows. As AI becomes embedded in every field, the ability to communicate precisely with these systems will be as essential as knowing how to write a clear email or conduct a search query.



💡 **The bottom line:** Your AI is only as smart as your prompt. Invest 30 extra seconds crafting a refined prompt and you'll save 30 minutes fixing a bad response.

## Final Learning Outcomes

01

Understood the six core prompting patterns and their optimal use cases

02

Empirically verified that structured prompts outperform vague prompts across all metrics

03

Developed practical skills in evaluating and scoring AI output quality

04

Recognised prompt engineering as a transferable, high-value professional competency

05

Gained appreciation for the iterative nature of prompt refinement – it's a craft, not a one-shot task